

Assignment III - PHYS 708

BegadE

October 2025

Problem 1

If the maximum energy of cosmic-ray protons is determined by the condition that the particle gyro-radius should be less than the size of the acceleration region, calculate the maximum energy of a proton for a spherical source of radius $R = 100$ parsec and a magnetic field strength of $B = 1 \mu\text{G}$. Take $\theta = 90^\circ$.

The gyroradius of a relativistic charged particle is given by

$$r = \frac{pc \sin \theta}{Ze Bc},$$

where p is the relativistic momentum, Z is the charge number, and B is the magnetic field strength.

For maximum confinement, we require the gyro-radius to be smaller than the source radius:

$$r \leq R_{\text{src}}.$$

Taking $\theta = 90^\circ$ i.e. $\sin \theta = 1$, the maximum rigidity (energy per charge) is

$$\mathcal{R}_{\text{max}} = BcR_{\text{src}}.$$

For a proton ($Z = 1$), the energy is related to the rigidity by $E \simeq pc \simeq e\mathcal{R}$, so

$$E_{\text{max}} = e B c R_{\text{src}}.$$

Substitute the given values:

$$B = 1 \mu\text{G} = 10^{-10} \text{ T},$$

$$R_{\text{src}} = 100 \text{ pc} = 100 \times 3.086 \times 10^{16} \text{ m} = 3.086 \times 10^{18} \text{ m},$$

$$c = 2.998 \times 10^8 \text{ m s}^{-1},$$

$$e = 1.602 \times 10^{-19} \text{ C}.$$

Thus,

$$\begin{aligned} E_{\text{max}} &= (1.602 \times 10^{-19})(10^{-10})(2.998 \times 10^8)(3.086 \times 10^{18}) \text{ J} \\ &\simeq 1.48 \times 10^{-2} \text{ J}. \end{aligned}$$

Converting to electronvolts:

$$E_{\text{max}} = \frac{1.48 \times 10^{-2}}{1.602 \times 10^{-19}} \text{ eV} \simeq 9.3 \times 10^{16} \text{ eV}.$$

$$E_{\text{max}} \approx 9 \times 10^{16} \text{ eV} \approx 90 \text{ PeV}.$$

Conclusion

The maximum energy a proton can be accelerated to in a region of radius 100 pc and magnetic field $1 \mu\text{G}$ is approximately 9×10^{16} eV.

Problem 2

Muons at rest have a mean lifetime (decay time) of 2.2×10^{-6} seconds. Calculate the relativistic lifetime of a muon of kinetic energy 1 GeV produced from cosmic-ray interaction in the Earth's atmosphere. Compare the relativistic lifetime with the time the muon takes to reach the ground if it is produced at a height of 5 km above the ground. Assume that the muon travels vertically downward.

Given:

Proper (rest) lifetime: $\tau_0 = 2.2 \times 10^{-6}$ s
Muon rest energy: $m_\mu c^2 = 105.66$ MeV
Kinetic energy: $K = 1\text{GeV} = 1000$ MeV
Production height: $h = 5.0$ km
Speed of light: $c = 2.998 \times 10^8$ m s⁻¹

1. **We first find the Lorentz factor γ and speed factor β .**

Total energy:

$$E = K + m_\mu c^2 = 1000 + 105.66 = 1105.66 \text{ MeV.}$$

Hence

$$\gamma = \frac{E}{m_\mu c^2} = \frac{1105.66}{105.66} \simeq 10.46, \quad \beta = \sqrt{1 - \frac{1}{\gamma^2}} \simeq 0.9954.$$

2. **We can now calculate the relativistic (dilated) lifetime τ .**

$$\tau = \gamma \tau_0 = (10.46) (2.2 \times 10^{-6} \text{ s}) \simeq 2.30 \times 10^{-5} \text{ s} = 23.0 \mu\text{s}.$$

3. **We now check the flight time for a muon to travel 5 km.**

$$t_{\text{flight}} = \frac{h}{\beta c} = \frac{5.0 \times 10^3}{(0.9954)(2.998 \times 10^8)} \simeq 1.68 \times 10^{-5} \text{ s} = 16.8 \mu\text{s}.$$

4. **Comparing both τ and t_{flight} yields**

$$\tau \approx 23.0 \mu\text{s}, \quad t_{\text{flight}} \approx 16.8 \mu\text{s}.$$

Since $t_{\text{flight}} < \tau$, many muons reach the ground.

We can also calculate the decay length for the muon to study how far a muon can travel before decaying.

$$\ell = \beta c \tau = (0.9954) (2.998 \times 10^8) (2.30 \times 10^{-5}) \simeq 6.9 \text{ km}.$$

Conclusion. A 1 GeV atmospheric muon has $\gamma \approx 10.46$, $\beta \approx 0.995$, a dilated lifetime of $\approx 23 \mu\text{s}$, and requires $\approx 16.8 \mu\text{s}$ to fall 5 km. Therefore, we can conclude that many muons reach the ground.

Problem 3

Using the expression for the emissivity of gamma rays from π^0 -decay, show that the gamma-ray spectrum follows a power-law similar to that of the cosmic-ray protons at very high energies.

Goal

Starting from the emissivity expression for γ rays from π^0 decay, show that if the parent proton spectrum is a power law $J_p(E_p) = J_0 E_p^{-\Gamma}$, then the resulting γ -ray spectrum obeys the same power law for higher energies.

$$q_\gamma^{\pi^0}(E_\gamma) \propto E_\gamma^{-\Gamma}$$

Solution

1. Stating the emissivity of γ -rays

$$q_\gamma^{\pi^0}(E_\gamma) = 8\pi n_H \int_{E_{\pi,\min}(E_\gamma)}^{\infty} \frac{dE_\pi}{\sqrt{E_\pi^2 - m_{\pi^0}^2 c^4}} \int_{E_{p,\min}(E_\pi)}^{\infty} dE_p J_p(E_p) \frac{d\sigma_{\pi^0}(E_\pi, E_p)}{dE_\pi}.$$

2. Working with the π^0 production term

At high energies, the produced π^0 carries on average a fixed fraction k_π of the proton energy, where the differential cross-section can be approximated by:

$$\frac{d\sigma_{\pi^0}}{dE_\pi} \approx \sigma_{pp}(E_p) \delta(E_\pi - k_\pi E_p).$$

Insert this in the inner integral:

$$\int dE_p J_p(E_p) \sigma_{pp}(E_p) \delta(E_\pi - k_\pi E_p) = \int dE_p J_p(E_p) \sigma_{pp}(E_p) \delta\left(k_\pi [E_p - E_\pi/k_\pi]\right)$$

and use $\delta(ax) = \delta(x)/|a|$ to obtain

$$= \frac{1}{k_\pi} \int dE_p J_p(E_p) \sigma_{pp}(E_p) \delta\left(E_p - \frac{E_\pi}{k_\pi}\right) = \frac{1}{k_\pi} J_p\left(\frac{E_\pi}{k_\pi}\right) \sigma_{pp}\left(\frac{E_\pi}{k_\pi}\right).$$

3. Working with the γ -ray production term

For high energies such that $E_\pi \gg m_{\pi^0} c^2$, $\sqrt{E_\pi^2 - m_{\pi^0}^2 c^4} \simeq E_\pi$. Since σ_{pp} varies slowly with energy at high energies, we can consider it as a constant that does not affect the spectral *shape*. Thus

$$q_\gamma^{\pi^0}(E_\gamma) \propto \int_{E_{\pi,\min}(E_\gamma)}^{\infty} \frac{dE_\pi}{E_\pi} J_p\left(\frac{E_\pi}{k_\pi}\right).$$

4. Insert the proton spectrum power law in the γ -ray production term

$$J_p\left(\frac{E_\pi}{k_\pi}\right) = J_0 \left(\frac{E_\pi}{k_\pi}\right)^{-\Gamma} = J_0 k_\pi^\Gamma E_\pi^{-\Gamma}.$$

Therefore

$$q_\gamma^{\pi^0}(E_\gamma) \propto \int_{E_{\pi,\min}(E_\gamma)}^{\infty} \frac{E_\pi^{-\Gamma}}{E_\pi} dE_\pi = \int_{E_{\pi,\min}(E_\gamma)}^{\infty} E_\pi^{-(\Gamma+1)} dE_\pi = \frac{1}{\Gamma} E_{\pi,\min}(E_\gamma)^{-\Gamma}.$$

5. Relate pion and photon energies

From $\pi^0 \rightarrow 2\gamma$ kinematics,

$$E_{\pi,\min}(E_\gamma) \xrightarrow{E_\gamma \gg m_{\pi^0} c^2/2} 2E_\gamma.$$

Conclusion We can finally conclude that

$$q_{\gamma}^{\pi^0}(E_{\gamma}) \propto (2E_{\gamma})^{-\Gamma} \propto E_{\gamma}^{-\Gamma},$$

i.e., at high energies the γ -ray spectrum produced by π^0 decay *obeys the same power-law index* as the parent cosmic-ray proton spectrum.

Problem 4

If the energy loss rate of a high-energy electron due to synchrotron losses is given by

$$\frac{dE}{dt} = -aE^2,$$

where E is the electron energy and a is a positive constant, show that the lifetime of the electron is proportional to $1/E$.

Solution. Separate variables and integrate:

$$\frac{dE}{E^2} = -a dt \implies \int_{E_0}^{E(t)} E^{-2} dE = -a \int_0^t dt.$$

This yields

$$-\frac{1}{E(t)} + \frac{1}{E_0} = -at \implies \frac{1}{E(t)} = \frac{1}{E_0} + at, \quad t = \frac{1}{a} \left(\frac{1}{E(t)} - \frac{1}{E_0} \right).$$

Define the cooling time to go from E_i to E_f by

$$t_{\text{cool}}(E_i \rightarrow E_f) = \frac{1}{a} \left(\frac{1}{E_f} - \frac{1}{E_i} \right).$$

If $E_f \ll E_i$, then

$$t_{\text{cool}} \approx \frac{1}{aE_i},$$

so the lifetime scales as $t \propto 1/E$.

Equivalently, the instantaneous loss timescale $t_{\text{loss}} \equiv E/|dE/dt|$ gives

$$t_{\text{loss}} = \frac{E}{aE^2} = \frac{1}{aE} \propto \frac{1}{E}.$$